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REDUCED ORDER MODELING OF
TURBULENT FLOWS USING
 β -VARIATIONAL AUTOENCODERS

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Abstract

This project investigates and compares linear and non-linear data-driven reduced-order modeling (ROM) approaches for turbulent fluid dynamics: Proper Orthogonal Decomposition (POD) and the β -Variational Autoencoder (β -VAE). Both methodologies are applied to three distinct flow regimes—a laminar two-dimensional cylinder wake ($Re = 100$), a wall-bounded turbulent channel flow ($Re_\tau = 1000$), and homogeneous isotropic box turbulence ($Re_\lambda \approx 433$)—to evaluate their capacity to compress spatiotemporal kinematics onto low-dimensional manifolds while preserving underlying Navier-Stokes physics. Quantitative analysis reveals that while POD remains computationally optimal for simple, linear limit cycles (capturing 98.5% of energy at rank 3 for the cylinder wake), the β -VAE demonstrates strictly superior compression efficiency in highly non-linear regimes. For the turbulent channel flow, the β -VAE captures 96.0% of the kinetic energy at a latent dimension of $z = 20$ compared to 60.0% for POD. Similarly, in the isotropic turbulence case, the β -VAE captures 94.0% of the kinetic energy at a latent dimension of $z = 10$, compared to 79.5% for POD, while successfully preserving the $-5/3$ Kolmogorov inertial subrange and non-Gaussian statistical intermittency. Furthermore, a Transformer architecture trained on the β -VAE latent codes was explored for short-horizon temporal forecasting of the chaotic regimes, though it was ultimately constrained by dataset sparsity.



1 Introduction

Turbulent fluid dynamics are fundamentally governed by the non-linear convective acceleration within the Navier-Stokes equations, generating a continuous cascade of interacting spatiotemporal scales. Because the computational degrees of freedom required to fully resolve a turbulent flow scale with the Reynolds number as $O(Re^{9/4})$ [11], Direct Numerical Simulation (DNS) remains prohibitively expensive for most high-Reynolds-number engineering applications. To bridge this gap, Reduced-Order Models (ROMs) are employed to project the essential kinematics of the high-dimensional flow onto a low-dimensional subspace, enabling accelerated analysis, optimization, and control.

The most classical and widely utilized ROM technique is Proper Orthogonal Decomposition (POD). Often computed via the Singular Value Decomposition (SVD), POD constructs an optimal linear orthogonal basis that strictly maximizes the captured turbulent kinetic energy per mode [15]. When a flow is dominated by macroscopic, periodic coherent structures—such as the von Kármán vortex shedding in a low-Reynolds cylinder wake—POD is highly efficient. However, as the flow transitions to fully turbulent regimes characterized by broadband energy spectra and localized, intermittent events (e.g., near-wall sweeps and ejections), the assumption of global linear superposition becomes a critical mathematical bottleneck. Consequently, the energy spectrum of the POD modes exhibits a notoriously slow decay, requiring a computationally cumbersome number of modes to reach an acceptable reconstruction accuracy.

Recent paradigm shifts in deep learning have motivated the deployment of non-linear convolutional autoencoders as a superior alternative to linear POD [4, 14, 6, 17]. Unlike standard autoencoders, the β -Variational Autoencoder (β -VAE) learns a probabilistic, non-linear mapping from the high-dimensional spatial state to a highly compressed latent space. The distinguishing feature of the β -VAE is its heavily penalized Kullback-Leibler (KL) divergence term, which enforces structural independence within the latent code. As demonstrated by Solera-Rico et al. [14], β -VAEs trained on turbulent flow snapshots can achieve significantly higher compression ratios than POD at equivalent reconstruction accuracies, while simultaneously producing disentangled latent representations where individual dimensions map directly to independent physical mechanisms.

This project implements the non-linear ROM framework proposed by Solera-Rico et al. [14] entirely from scratch within the MATLAB environment. To rigorously evaluate the scalability and compression efficiency of linear versus non-linear manifolds, the algorithms are applied to three distinct flow configurations of increasing turbulent complexity: a laminar, two-dimensional cylinder wake limit cycle ($Re = 100$); an anisotropic, wall-bounded turbulent channel flow ($Re_\tau = 1000$); and highly chaotic, homogeneous isotropic box turbulence ($Re_\lambda \approx 433$). Furthermore, to model the temporal evolution of these highly compressed representations, a sequence-to-sequence Transformer encoder architecture is trained on the β -VAE latent time series. While the Transformer successfully mitigates the vanishing gradient problem inherent in traditional recurrent networks, its application for short-horizon forecasting in the turbulent cases was ultimately constrained by the temporal sparsity of the available snapshot data.

The report is organized as follows. Section 2 details the mathematical foundation and architecture of POD, the β -VAE, and the Transformer network. Section 3 characterizes the three underlying flow configurations and their corresponding DNS datasets. Section 4 presents the quantitative compression and reconstruction results for each flow regime. Section 5 critically discusses the comparative strengths and theoretical limitations of each methodology, and Section 6 summarizes the primary findings of the study.



2 Methodology

The development of a Reduced-Order Model (ROM) for turbulent flows requires a two-stage framework: spatial dimensionality reduction to extract the dominant coherent structures, and temporal sequence modeling to predict the evolution of these structures over time.

2.1 Proper Orthogonal Decomposition

POD seeks an orthonormal basis $\{\mathbf{U}_i\}_{i=1}^r$ in which the projection of the fluctuating velocity field captures the maximum possible energy. Given a data matrix $\mathbf{X} \in \mathbb{R}^{N \times M}$ whose columns are the $M = N_t$ mean-subtracted flow snapshots vectorized over all N spatial degrees of freedom, POD is computed via the truncated Singular Value Decomposition (SVD):

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top \quad \text{and the truncated SVD,} \quad \mathbf{X}_r = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^\top \quad (1)$$

where $\mathbf{U}_r \in \mathbb{R}^{N \times r}$ contains the spatial modes (left singular vectors), $\mathbf{\Sigma}_r = \text{diag}(\sigma_1, \dots, \sigma_r)$ the singular values in descending order, and $\mathbf{V}_r \in \mathbb{R}^{M \times r}$ the temporal coefficients (right singular vectors). The temporal coefficient for mode i at time t is $a_i(t) = \sigma_i V_{ti}$.

A rank- r POD reconstruction of snapshot $\mathbf{x}(t)$ is obtained as

$$\hat{\mathbf{x}}(t) = \bar{\mathbf{x}} + \sum_{i=1}^r a_i(t) \mathbf{U}_i \quad (2)$$

where $\bar{\mathbf{x}}$ is the time-mean field.

Reconstruction quality is measured by the percentage of captured energy [14, 4]

$$E = \left(1 - \left\langle \frac{\sum_{i=1}^N (u'_i - \hat{u}'_i)^2}{\sum_{i=1}^N u_i'^2} \right\rangle \right) \times 100\% \quad (3)$$

where N is the number of grid points, u'_i is the velocity fluctuation value and $\hat{u}'_i$ is its low-order reconstruction.

While mathematically optimal in the L_2 norm for linear representations, POD relies strictly on linear superposition. Because the Navier-Stokes equations are inherently non-linear due to the convective acceleration term $(\mathbf{u} \cdot \nabla) \mathbf{u}$, POD struggles to efficiently represent traveling waves or highly localized intermittent structures (such as near-wall sweeps and ejections). Consequently, POD acts as a linear low-pass filter, often requiring a prohibitively large number of modes to capture the dissipative micro-scales of the turbulent energy cascade.

2.2 β - Variational Autoencoder

To overcome the limitations of linear hyperplanes, we utilize deep learning to map the turbulence onto a non-linear, low-dimensional manifold. A Variational Autoencoder (VAE) is a neural network that simultaneously learns a probabilistic encoder $q_\Phi(\mathbf{z}|\mathbf{x})$ and a decoder $p_\theta(\mathbf{x}|\mathbf{z})$, where $\mathbf{x} \in \mathbb{R}^N$ is a flow snapshot and $\mathbf{z} \in \mathbb{R}^d$ is the latent code. The encoder outputs the parameters (μ, σ^2) of a Gaussian, and samples are drawn via the reparameterization trick $\mathbf{z} = \mu + \sigma \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ [4].

The β -VAE loss function is expressed as a balance between a reconstruction term and a regularization

term: [4]

$$\mathcal{L} = \underbrace{\|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2}_{\text{reconstruction}} + \beta \underbrace{D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{regularization}} \quad (4)$$

where $\tilde{\mathbf{x}}$ is the reconstructed input, $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the prior distribution, and D_{KL} is the Kullback-Leibler divergence. The hyperparameter β heavily penalizes correlations within the latent space, promoting physical disentanglement. In the context of fluid flows, disentanglement means that each z_i ideally corresponds to an independent kinematic mechanism (e.g., vortex shedding frequency, rotational amplitude, or shear layer thickness), preventing the "mode mixing" commonly observed in standard autoencoders.

The encoder-decoder architecture used in this work follows Solera-Rico et al. [14] and consists of convolutional layers in the encoder and mirrored convolutional layers in the decoder, operating on 2D flow snapshots of shape $[N_y, N_x, C]$, where C is the number of velocity components. Convolutional Neural Networks (CNNs) are uniquely suited for this task due to their translation invariance, allowing them to capture spatial hierarchies from the large integral scales down to the inertial subrange. The encoder maps each snapshot to a latent vector $\mathbf{z}$. Reconstruction is obtained by passing $\mathbf{z}$ through the decoder and denormalizing with the training set mean and standard deviation.

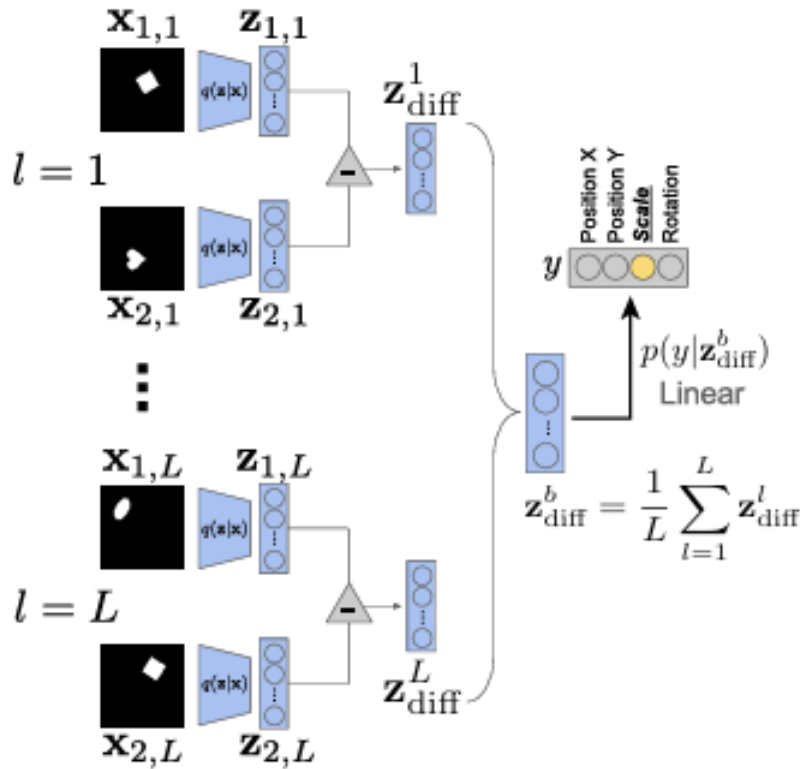


Figure 1: Schematic of the disentanglement metric proposed for the β -VAE [7]. By evaluating the average pairwise difference ($\mathbf{z}_{\text{diff}}^b$) across a batch of samples where one target generative factor y is held constant, the network forces the latent space to map independent physical mechanisms (e.g., vortex shedding frequency or amplitude) to distinct, orthogonal latent dimensions.

The β -VAE modes are defined as the decoder response to unit perturbations in latent space:

$$\mathbf{U}_i^{\beta\text{-VAE}} = D(\mathbf{z}_i) - D(\mathbf{0}) \quad (5)$$

where $\mathbf{z}_i$ is the i -th latent space vector and $D(\cdot)$ denotes the decoder. This mode definition makes the β -

VAE modes directly comparable to POD modes as spatial structures that can be summed onto a mean field.

The training hyperparameters for used for all the cases are the following: learning rate 3×10^{-4} (Adam optimizer), batch size 32, up to 500 epochs and $\beta = 5 \times 10^{-4}$.

2.3 Temporal Modeling via Transformers

While the β -VAE successfully compresses the spatial grid, fluid dynamics are inherently spatiotemporal. To model the temporal evolution of the flow without solving the full Navier-Stokes equations, we predict the future trajectory of the highly compressed latent vectors $\mathbf{z}(t)$ over time.

Traditionally, Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks are used for sequence modeling. However, turbulent flows exhibit complex, long-range temporal dependencies (such as the low-frequency "flapping" of a wake) that trigger the vanishing gradient problem in sequential networks. To address this, we utilize a Transformer architecture [16], which entirely discards recurrence in favor of the Multi-Head Self-Attention mechanism.

Given a temporal sequence of past latent states $\mathbf{Z}_{in} = [\mathbf{z}_{t-k}, \dots, \mathbf{z}_t]$, the Transformer learns to predict the subsequent states. The core of the architecture is the scaled dot-product attention, defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V} \quad (6)$$

where $\mathbf{Q}$ (Queries), $\mathbf{K}$ (Keys), and $\mathbf{V}$ (Values) are linear projections of the input latent sequence, and d_k is the scaling dimension. In a fluid dynamics context, the attention mechanism calculates an "attention score" between every pair of time steps simultaneously. This allows the model to explicitly map how a turbulent event at $t - 10$ (e.g., vortex detachment) mathematically influences the flow field at $t + 1$, bypassing intermediate temporal bottlenecks.

By coupling the non-linear spatial compression of the β -VAE with the parallelized, long-range temporal forecasting of the Transformer, we construct a fully data-driven ROM capable of predicting chaotic fluid kinematics at a fraction of the computational cost of traditional CFD.

3 Datasets and Flow Configurations

3.1 Two-Dimensional Cylinder Wake

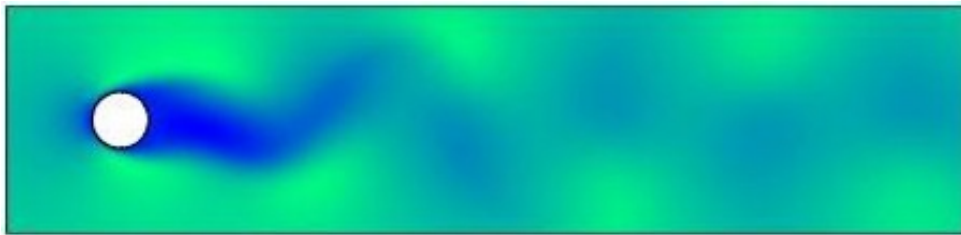


Figure 2: Instantaneous velocity field of the 2D laminar cylinder wake at $Re = 100$. The direct numerical simulation was conducted using the Nek5000 spectral element solver [2].

The cylinder flow dataset consists of 2D snapshots [2] of the flow around a circular cylinder. Reynolds numbers considered is $Re = 100$. The spatial domain is discretized on a 500×500 grid and the dataset

contains $N_t = 500$ snapshots per case. The two velocity components (u, v) are stored in a tensor of shape $[2, N_y, N_x, N_t]$. At these Reynolds numbers the flow is periodic with a well-defined shedding frequency, making the cylinder case an ideal validation benchmark.

The physics of flow past a circular cylinder is heavily dependent on the Reynolds number. At $Re = 100$, the flow resides in the laminar periodic shedding regime, characterized by the formation of the classic von Kármán vortex street [18]. In this state, the wake exhibits highly stable, alternating vortex shedding driven by a global absolute instability. Because the flow dynamics act as a low-dimensional limit cycle, the turbulent kinetic energy is concentrated within a few dominant frequencies. This dataset was selected to serve as a foundational baseline; the strictly periodic nature allows linear temporal algorithms, such as Dynamic Mode Decomposition (DMD), to effectively isolate the fundamental shedding frequency (Strouhal number) and structural growth rates without interference from chaotic broadband noise [12]. It allows for the rigorous mathematical validation of both the spatial and temporal reduction models before applying them to fully turbulent, non-linear regimes.

3.2 Turbulent Channel Flow

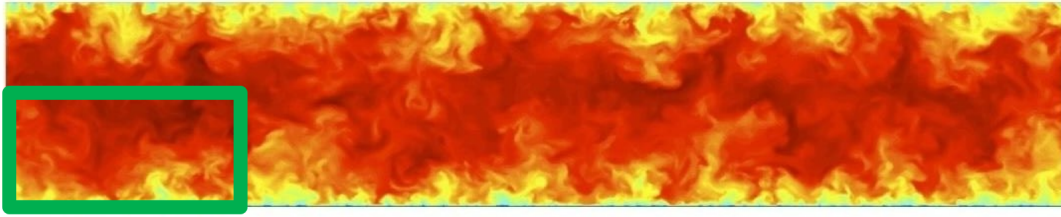


Figure 3: Cross-section of the streamwise velocity field (u_1) in the fully developed turbulent channel flow DNS at $Re_\tau = 1000$ [10]. The highlighted green bounding box illustrates the specific localized spatial domain extracted for the reduced-order modeling pipeline.

The channel flow dataset [10] corresponds to a 3D simulation of a pressure-driven flow between two parallel walls. The data is on a 256×256 grid with $N_t = 4000$ snapshots and two in plane velocity components (u, v) . Only one section representing the half height of the channel is considered. The friction velocity Reynolds number ($Re_\tau = 1000$), the friction velocity ($u_\tau = 0.0499$) and the viscosity ($\nu = 5 \times 10^{-5}$) are used to characterize the flow. The channel flow presents a much wider range of spatial scales than the cylinder case, requiring significantly more modes for accurate reconstruction.

The raw data for this slice is extracted from the Johns Hopkins Turbulence Databases (JHTDB) wall-bounded channel flow Direct Numerical Simulation (DNS) [5]. The full DNS solves the Navier-Stokes equations using a wall-normal B-spline and spectral method on a massive $2048 \times 512 \times 1536$ grid over a domain of $8\pi \times 2 \times 3\pi$. Physically, wall-bounded turbulence introduces strong spatial anisotropy and extreme near-wall velocity gradients. The boundary layer is dominated by highly non-linear, intermittent bursting events—specifically, low-speed ejections and high-speed sweeps—which serve as the primary mechanisms for turbulent kinetic energy production and Reynolds shear stress ($\overline{u'v'}$) generation [11]. This dataset was explicitly chosen to evaluate the capability of Reduced-Order Models (ROMs) to resolve these sharp, localized momentum transfers. Linear modal decompositions, such as POD, typically fail in this regime by structurally "smearing" the sweeps and ejections, making this an excellent benchmark to prove the necessity of the non-linear spatial mapping capabilities of the β -VAE.

3.3 Isotropic Box Turbulence

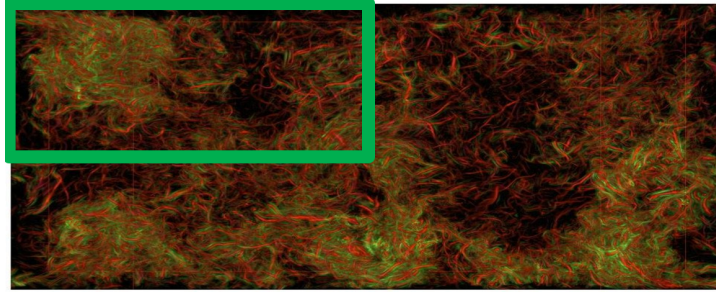


Figure 4: Visualization of the intense, multi-scale vorticity field ($\nabla \times \mathbf{u}$) within the forced homogeneous isotropic turbulence DNS at $Re_\lambda \approx 433$ [5]. The green bounding box denotes the extracted 2D planar slice utilized for evaluating ROM performance on statistically stationary turbulent chaos.

The forced isotropic turbulence dataset [5] represents homogeneous isotropic turbulence (HIT) in a box. The data is on a 256×256 grid at a wall-normal slice, with $N_t = 5024$ snapshots and three velocity components (u, v, w). This is the most demanding case in terms of ROM complexity. The flow lacks coherent structures and exhibits a broadband $k^{-5/3}$ energy spectrum across an extended inertial range. The box turbulence case is used to assess how well both methods preserve turbulence statistics (energy spectrum, velocity fluctuation PDF) beyond simple reconstruction accuracy.

This dataset is sourced from the JHTDB forced isotropic turbulence DNS, computed using a pseudo-spectral method on a 1024^3 grid within a $2\pi \times 2\pi \times 2\pi$ domain at a Taylor microscale Reynolds number of $Re_\lambda \approx 433$ [10]. To prevent the turbulence from naturally decaying, energy is artificially injected into the lowest wavenumbers to maintain a statistically stationary state. Homogeneous isotropic turbulence represents the purest form of fluid chaos, strictly governed by Kolmogorov's 1941 (K41) theory [8]. Energy cascades continuously from the large integral scales through the inertial subrange, eventually reaching the Kolmogorov microscale where viscous dissipation occurs into heat. Furthermore, the flow exhibits intense small-scale intermittency, leading to non-Gaussian, "heavy-tailed" probability density functions (PDFs) for velocity fluctuations. We selected this case as the ultimate stress test for dimensionality reduction. Because the flow entirely lacks macroscopic coherent boundaries or forcing geometries, it rigorously tests whether the ROM physically preserves the non-linear energy transfer to dissipative scales, rather than merely acting as an arbitrary mathematical low-pass filter.

4 Results

4.1 Cylinder Wake Flow

4.1.1 Velocity Field Reconstruction

Figure 5 shows a representation of snapshot 256 out of 500 of the streamwise (u) and crosswise (v) velocity fields: true, POD reconstruction (rank 2), and β -VAE reconstruction ($d = 2$). Both reconstructions recover the large-scale vortex shedding pattern accurately. The corresponding spatial error maps (difference between real and reconstructed data) are shown in Figure 6.

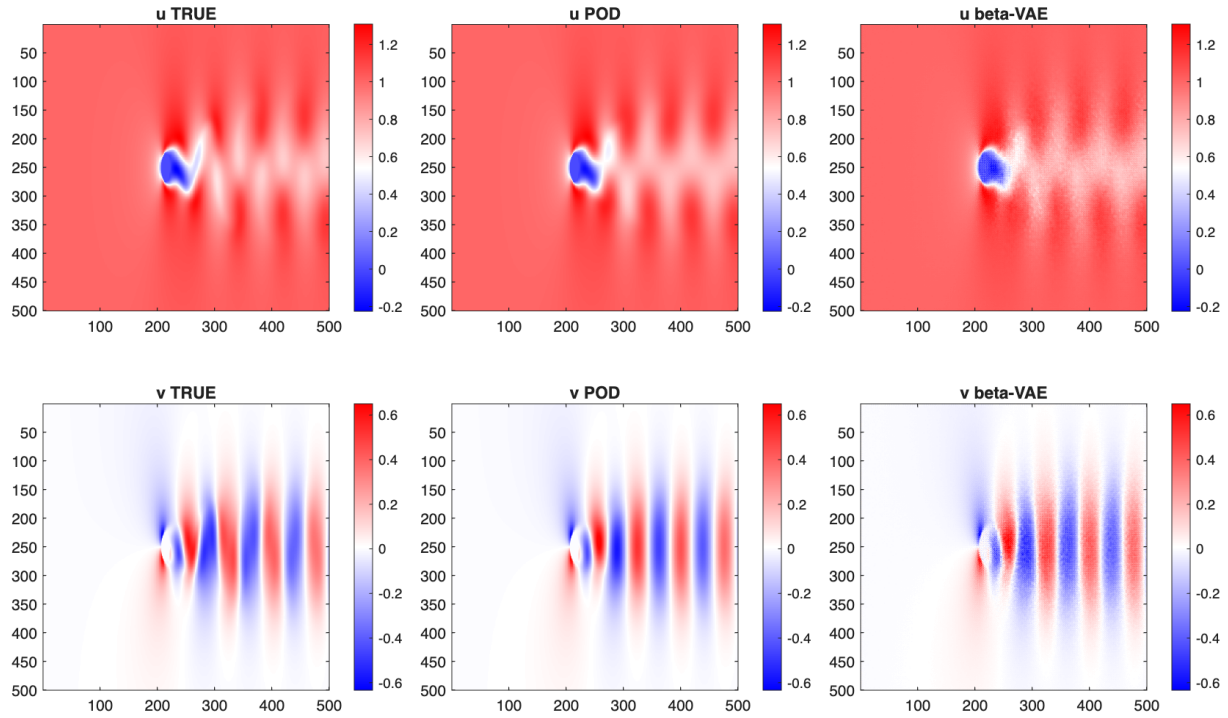


Figure 5: Velocity fields for the cylinder ($Re = 100$). Columns: true field, POD reconstruction ($r = 2$), β -VAE reconstruction ($d = 2$). Rows: u (top) and v (bottom).

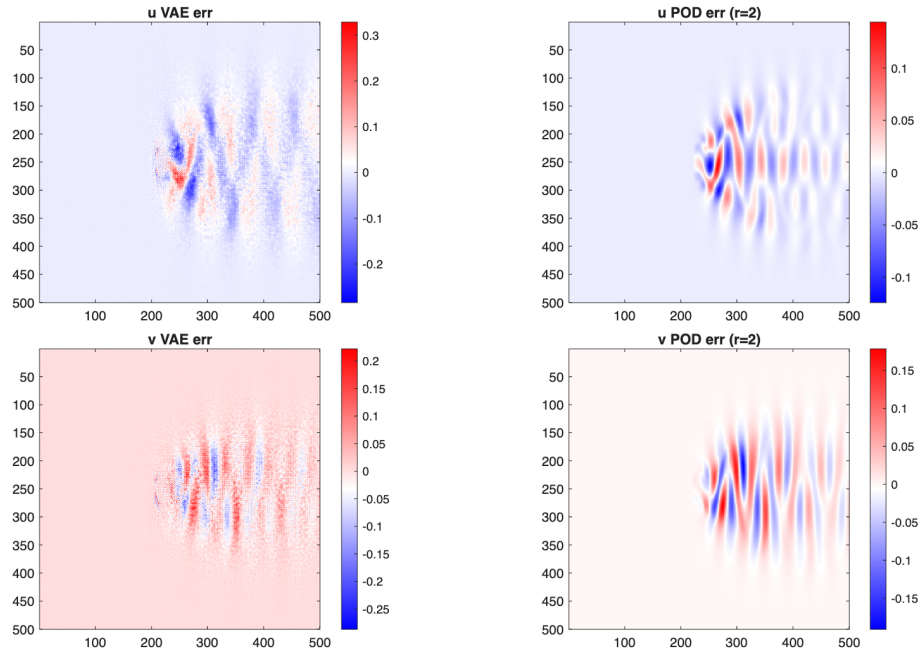
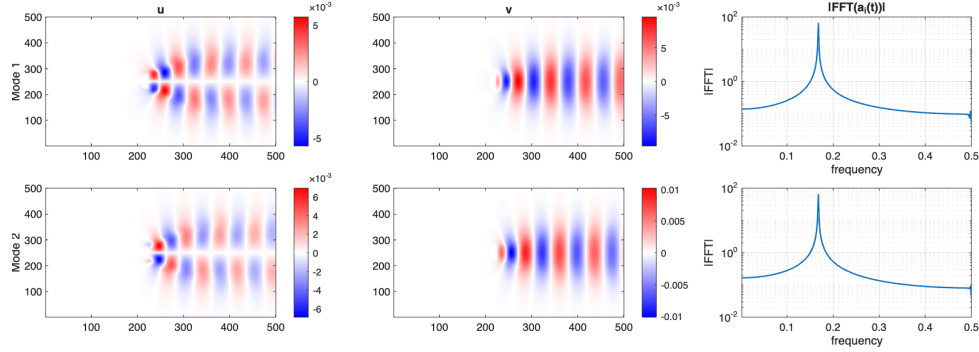


Figure 6: Error map ($\hat{u}' - u'_{\text{true}}$) for the cylinder flow. Left column: β -VAE ($d = 2$). Right column: POD ($r = 2$).

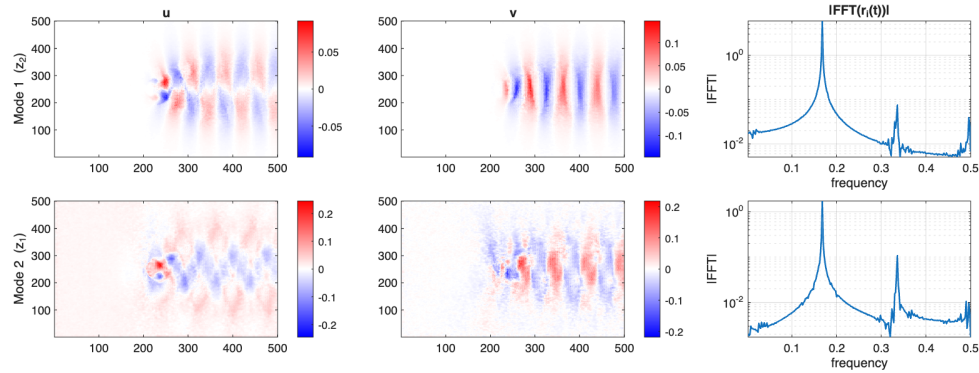
4.1.2 POD and β -VAE Modes

Figure 7 shows the first two POD spatial modes and their temporal spectra alongside the corresponding β -VAE modes with latent dimension of 2 ($d = 2$) for the cylinder case. Both methods identify the same dominant structure associated with the periodic Kármán vortex street. The temporal spectra of the POD

coefficients $a_i(t)$ reveal a sharp spectral peak at the dominant frequency, confirming the periodic nature of the flow. The β -VAE modes, defined by Eq.(5), exhibit similar spatial structures, validating the physical interpretability of the learned latent space. However, another frequency peak is clearly visible at higher frequency, indicating the representation of other physical phenomena not captured by the first two modes of the POD.



(a) POD spatial modes and temporal spectra.



(b) β -VAE modes ($d = 2$, Δ -mode definition).

Figure 7: First two spatial modes and temporal frequency spectra for the cylinder flow ($Re = 100$). Top: POD. Bottom: β -VAE with $d = 2$.

4.1.3 Reconstruction Accuracy

Figure 8 compares the captured energy as a function of the number of modes for POD and β -VAE at $d = 2$ and $d = 3$ on the cylinder case. POD captures approximately 49% of the fluctuation energy with a single mode, rising to $\sim 96\%$ with two modes. The β -VAE with $d = 2$ reaches a similar level ($\sim 96\%$ training, $\sim 95\%$ validation) with a latent dimension of 2, confirming that the nonlinear encoder is as compact as the optimal linear basis for this case. A β -VAE with $d = 3$ shows a marginal improvement, reaching $\sim 98\%$ with 3 modes.

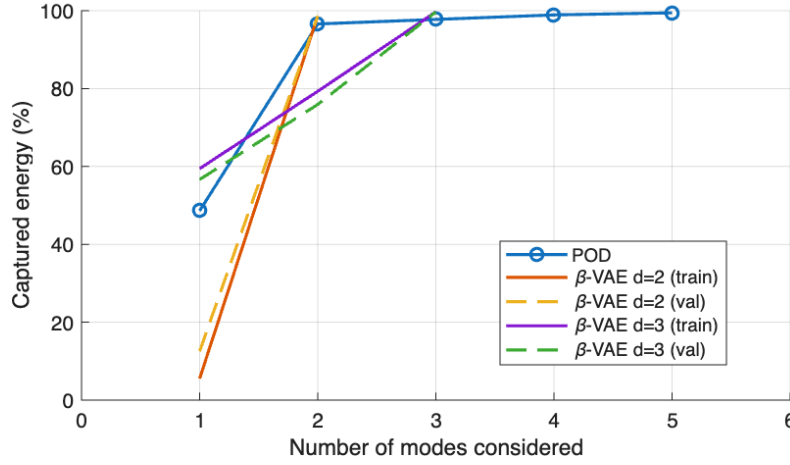


Figure 8: Captured energy (%) vs. number of modes for the cylinder flow ($Re = 100$). POD is compared with β -VAE at $d = 2$ and $d = 3$ (train and validation). Both methods require only 2 modes to capture $\sim 96\%$ of the fluctuation energy.

4.1.4 Latent Space Disentanglement

The structure of the latent representation is assessed through the determinant of the correlation matrix [4, 14]:

$$\det(\mathbf{R}) \quad R_{ij} = \frac{C_{ij}}{\sqrt{C_{ii}C_{jj}}} \quad (7)$$

where $\mathbf{R}$ is the correlation matrix and $\mathbf{C}_{ij}$ is the covariance matrix of the latent space where $1 \leq i \neq j \leq d$, where d is the latent space dimension. By construction the determinant of the correlation matrix of the POD is 1 meaning that the modes are orthogonal.

Figure 9 shows the correlation matrices $|R_{ij}|$ of the latent space for β -VAE ($\beta = 5 \times 10^{-4}$) and for the POD temporal coefficients at $Re = 100$. POD must be diagonal by construction but β -VAE exhibits a near diagonal structure. This indicates that both methods successfully identify the two components of the shedding cycle as independent directions in their respective latent spaces. The β -VAE achieves this disentanglement through the KL regularization, while POD achieves orthogonality by construction via the SVD.

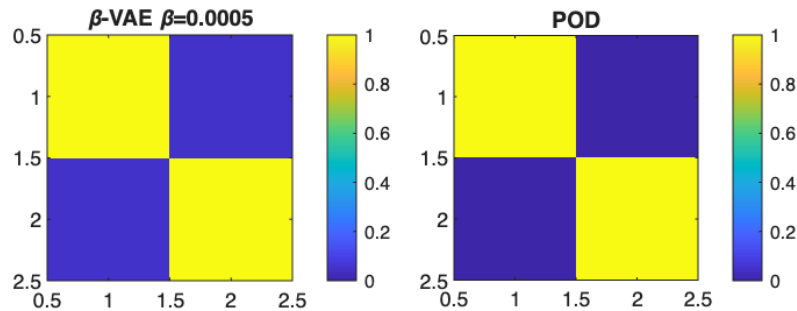


Figure 9: Absolute correlation matrices $|R_{ij}|$ of the latent representations for the cylinder flow ($Re = 100$). Left: β -VAE ($\beta = 5 \times 10^{-4}$, $d = 2$). Right: POD (2 modes). Both show near-zero cross-correlation between the two shedding modes.

4.2 Turbulent Channel Flow

4.2.1 Velocity Field Reconstruction

Figure 10 shows a reconstruction of snapshot 2000 out of 4000 for the channel flow, including the true field, the POD reconstruction and the β -VAE reconstruction. The β -VAE reconstruction captures the large-scale structures but also the smaller ones compared to POD. The reconstruction capability of the β -VAE is clearly visible in the crosswise component, where it is able to capture small structure but it filters out the smallest fine structures. On the other hand, the POD is only capturing large structures of the flow that are not close to the reality.

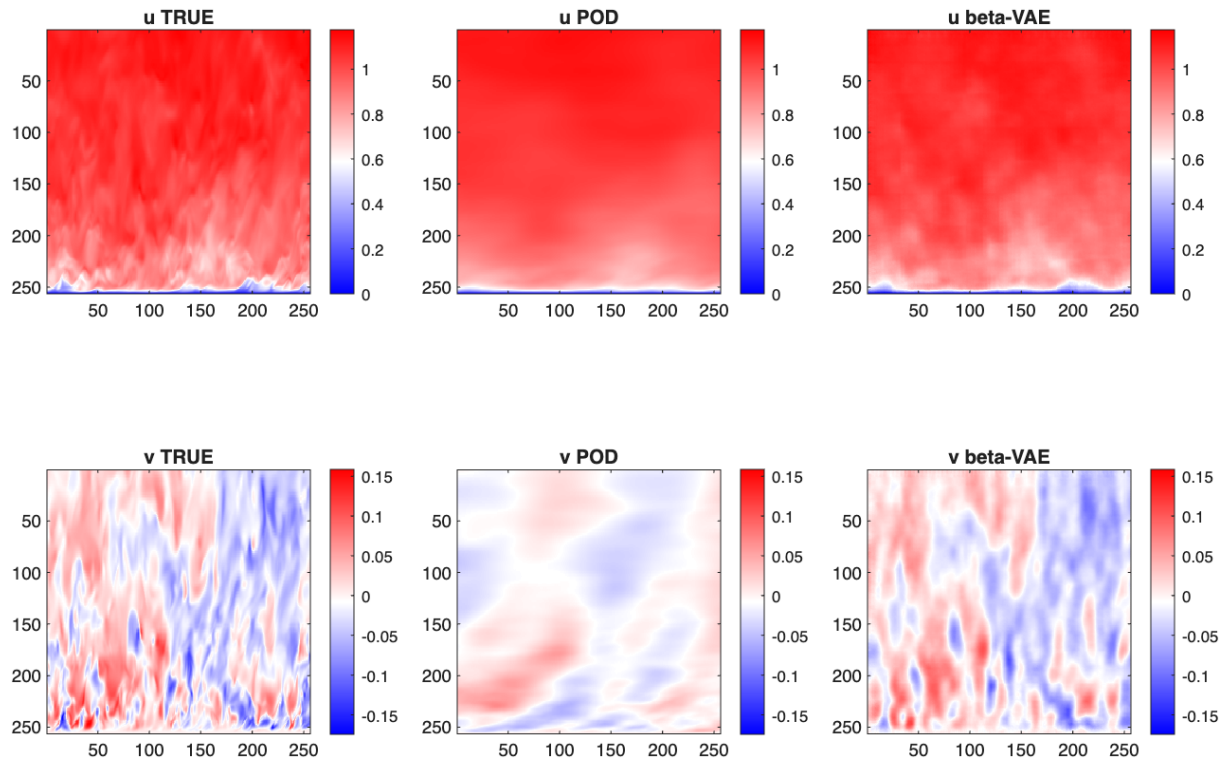
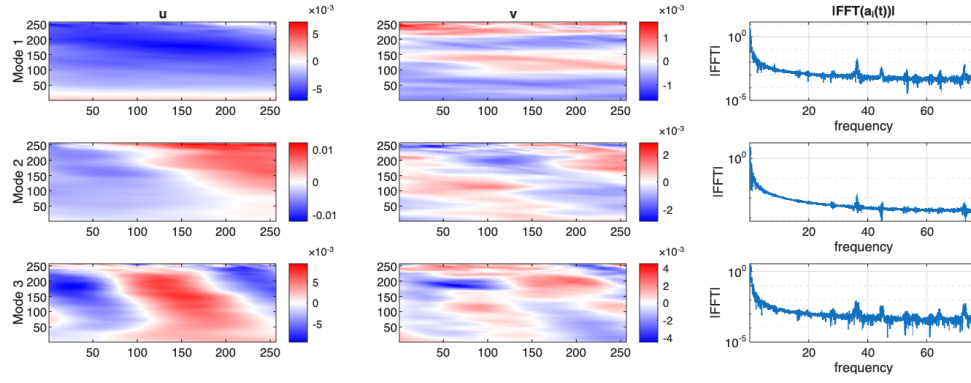


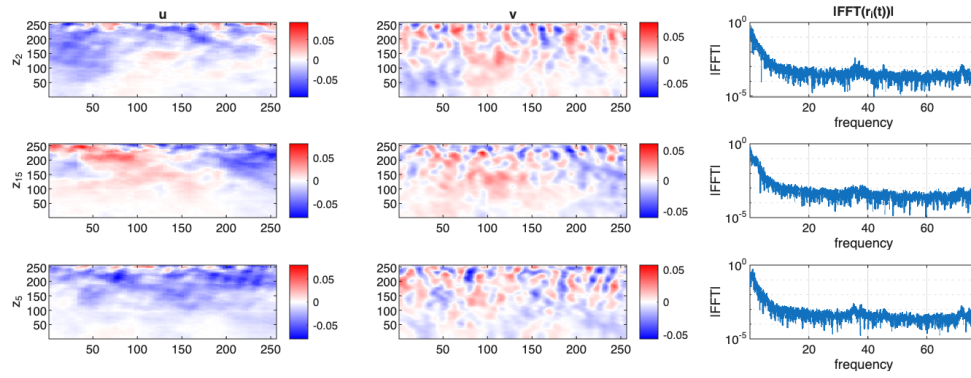
Figure 10: Velocity fields for the turbulent channel flow. True (left), POD reconstruction (center) and β -VAE reconstruction (right), for u (top row) and v (bottom row).

4.2.2 Mode Comparison

Figure 11 shows the leading POD modes and β -VAE modes for the channel flow. Unlike the cylinder case, the POD modes of the channel show large structures that don't represent periodic behavior of the flow. The β -VAE modes are qualitatively different: they represent nonlinear combinations of larger to smaller structures capturing multi-scale features within a single mode. The frequency analysis show no dominant frequency, instead a peak in 0 frequency that indicates no periodic behavior of the modes.



(a) POD spatial modes.



(b) β -VAE modes.

Figure 11: Spatial modes for the turbulent channel flow.

4.2.3 Energy Reconstruction

Figure 12 highlights the key advantage of the β -VAE for complex turbulent flows. POD requires approximately 75 modes to capture 80% of the fluctuation energy in the channel. In contrast, the β -VAE with $d = 20$ captures $\sim 95\%$ of the energy with only 20 latent dimensions (training set), and $d = 32$ shows marginal additional improvement. The gap between POD and β -VAE is most pronounced in the range $5 \leq r \leq 30$ modes, where the nonlinear encoder achieves compression ratios several times higher than the linear POD basis.

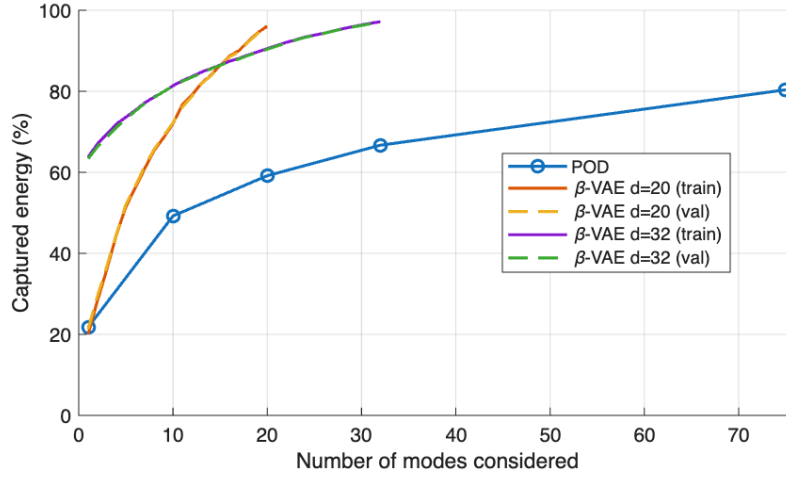


Figure 12: Captured energy (%) vs. number of modes for the turbulent channel flow.

4.2.4 Latent Space Disentanglement

Figure 13 presents the correlation matrices $|R_{ij}|$ of the latent space for the β -VAE ($\beta = 5 \times 10^{-4}$) and the POD temporal coefficients. The β -VAE correlation matrix exhibits a predominantly diagonal structure, with small off-diagonal values. This indicates that full orthogonality is not completely achieved, although the latent variables remain nearly uncorrelated. Similar behavior has also been reported in previous studies [4, 14, 17].

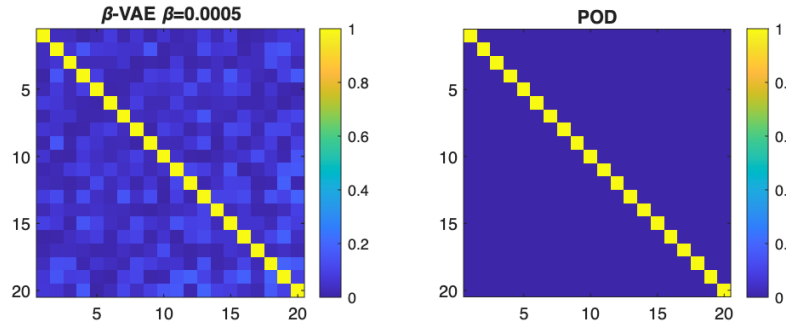


Figure 13: Absolute correlation matrices $|R_{ij}|$ of the latent representations for the channel flow.

4.2.5 Flow Characterization

The channel flow reconstructions are validated against theoretical expectations. Figure 14 shows the inner-scaled mean velocity profile $U^+(y^+)$ for the raw data and the two reconstructions. All three curves collapse onto the expected buffer and inertial layer, confirming that both POD and β -VAE preserve the mean flow statistics.

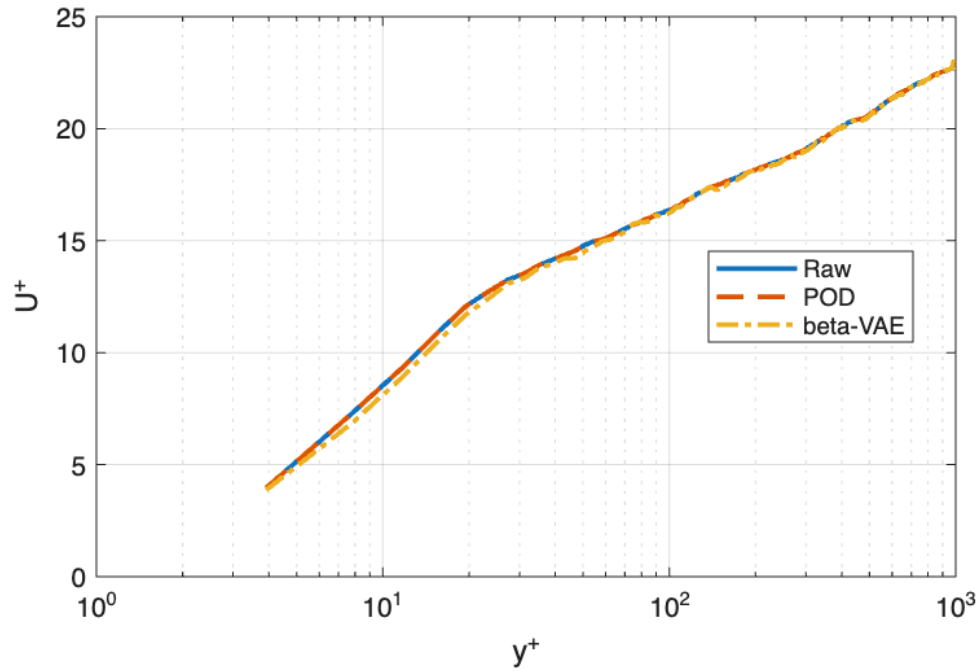


Figure 14: Inner-scaled mean velocity profile $U^+(y^+)$ for the turbulent channel flow. Raw DNS data (blue), POD reconstruction (orange dashed), and β -VAE reconstruction (yellow dash-dot) all follow the expected log-law scaling.

The channel flow reconstructions are further validated by examining the Reynolds shear stress in inner scaling. Figure 15 shows the profile of Reynolds stresses in inner scaling for the raw data and the two reconstructions. The raw data follows the expected trend, with a negative peak in the near-wall region and a gradual approach to zero towards the channel center.

The β -VAE reconstruction closely matches the raw data across most of the domain, accurately capturing both the magnitude and spatial distribution of the Reynolds shear stress. In contrast, the POD reconstruction exhibits significant deviations, particularly in the near-wall region, where it fails to reproduce the correct amplitude and even predicts non-physical positive values in parts of the domain.

These results indicate that, while both methods preserve mean flow statistics, the β -VAE provides an accurate representation of the turbulent statistics, highlighting its ability to capture nonlinear correlations associated with momentum transfer.

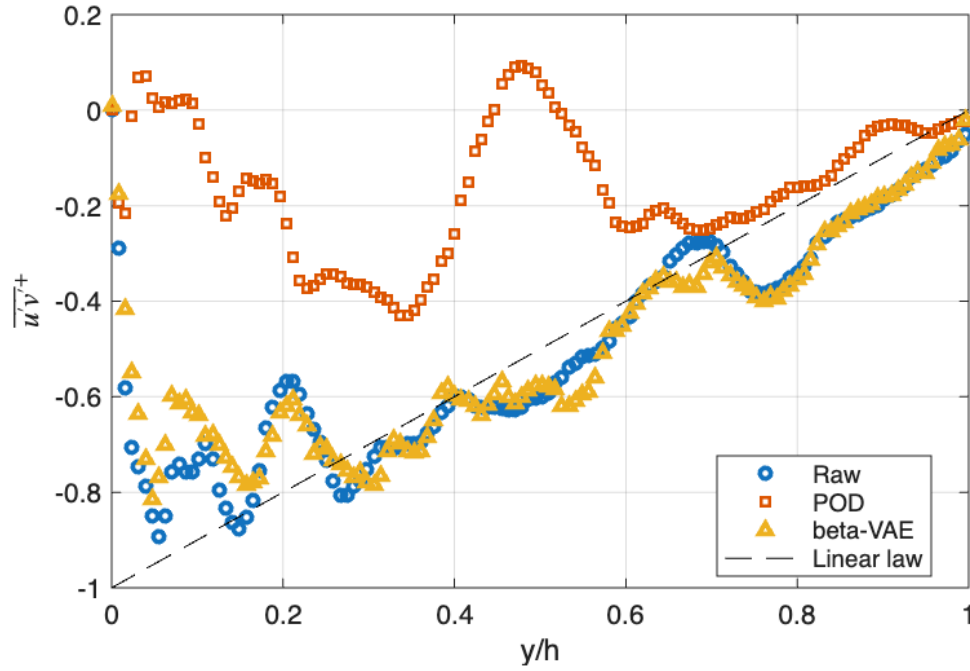


Figure 15: Inner-scaled Reynolds shear stress profile $\overline{u'v'}^+$ vs (y/h) for the turbulent channel flow. Raw DNS data (blue), POD reconstruction (orange dashed), and β -VAE reconstruction (yellow dash-dot). The β -VAE accurately reproduces the near-wall peak and overall trend, while POD shows significant deviations.

4.3 Isotropic Box Turbulence

4.3.1 Velocity Field Reconstruction

Figure 16 presents snapshot 2512 out of 5024 of the three velocity components (u , v , w) for the true data, POD reconstruction ($r = 5$), and β -VAE reconstruction ($d = 5$). The β -VAE reconstruction recovers finer-scale structures than POD at the same number of modes, consistent with the higher captured energy reported in Figure 18. Both methods smooth out the smallest-scale features, which represent the high-wavenumber part of the energy cascade.

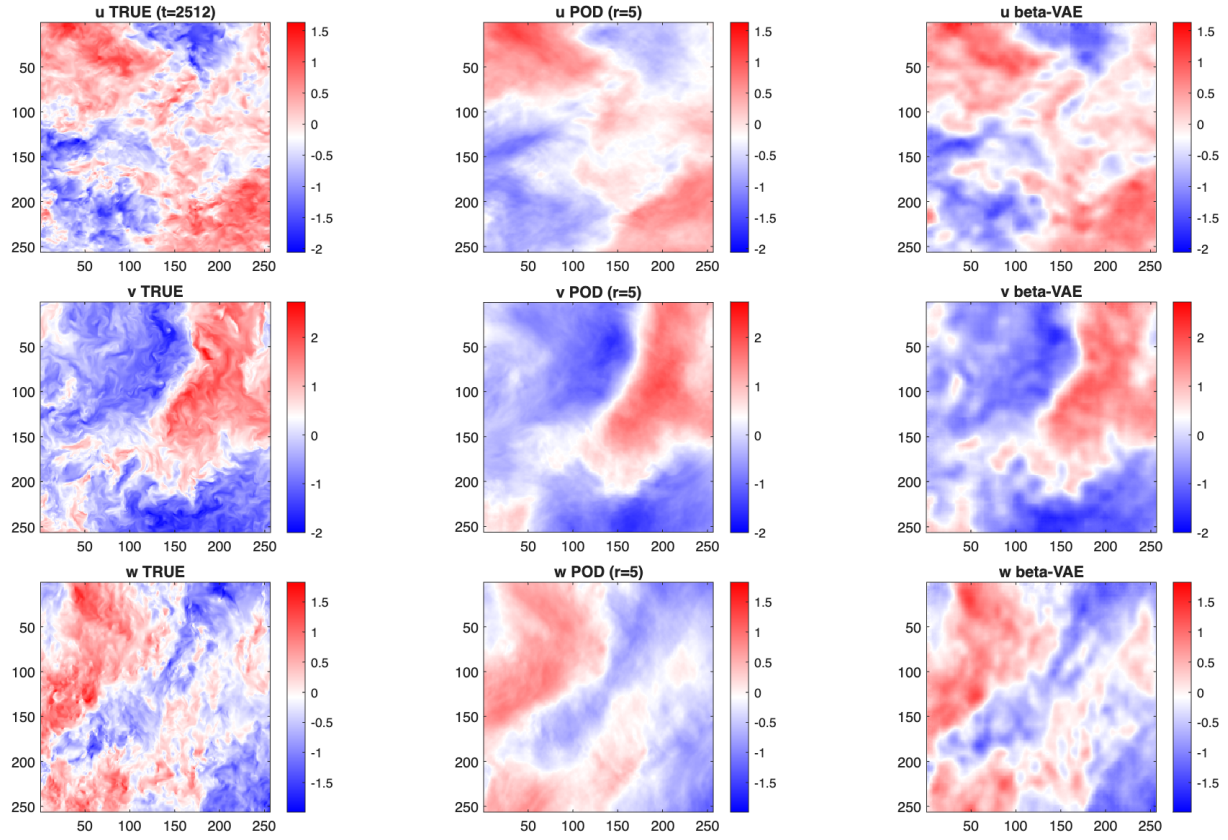
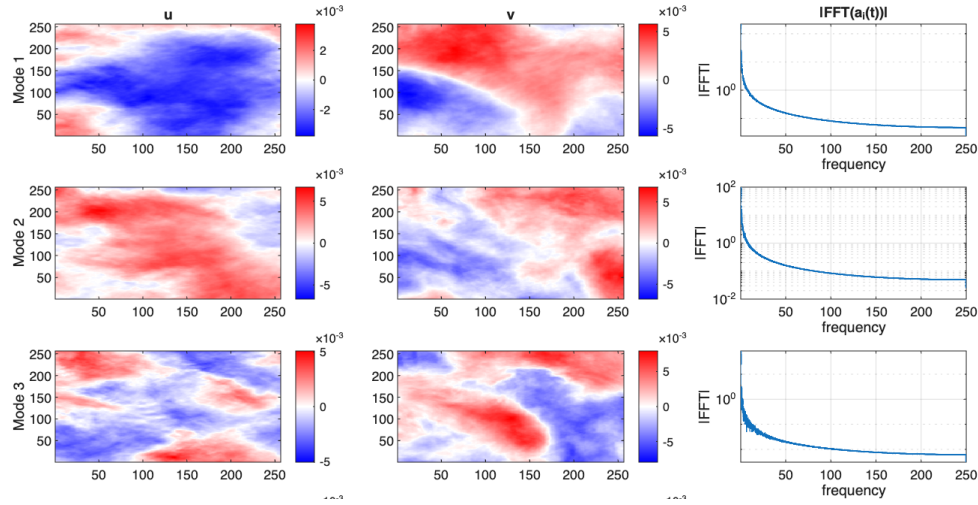


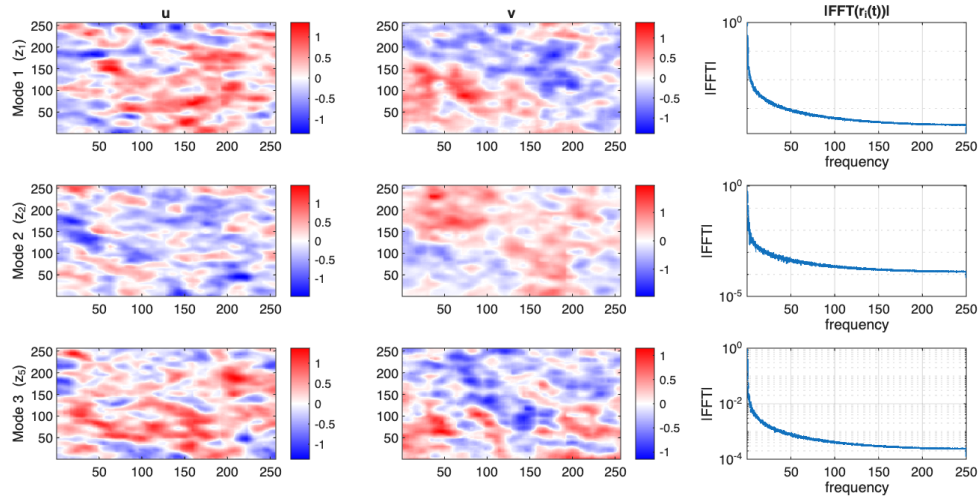
Figure 16: Velocity fields for the homogeneous isotropic turbulence case at a representative snapshot. Columns: true field, POD ($r = 5$), β -VAE ($d = 5$). Rows: u , v , w velocity components.

4.3.2 Mode Comparison

Figure 17 shows the leading POD and β -VAE modes for the box turbulence case. The POD modes show alternating positive/negative large patterns that don't represent any periodic pattern as seen in the Fourier transform. On the other hand, the β -VAE modes exhibit spatially complex multi-scale structures. Both Fourier transforms don't exhibit any dominant frequency. In this case, the flow is governed by stochastic dynamics, which is characteristic of homogeneous isotropic turbulence.



(a) Leading POD spatial modes for u , v , w .



(b) β -VAE modes (most energetic latent dims).

Figure 17: Spatial modes for the isotropic box turbulence case. Top: POD. Bottom: β -VAE.

4.3.3 Energy Reconstruction

Figure 18 shows the energy-vs-modes comparison for box turbulence. POD reaches approximately 85% captured energy at 20 modes. The β -VAE with $d = 5$ reaches $\sim 93\%$ training energy with only 5 latent dimensions, and $d = 10$ reaches $\sim 95\%$ with 10. This demonstrates that even for fully turbulent, structureless flows without dominant coherent motions, the nonlinear autoencoder significantly outperforms the linear POD in terms of energy compression efficiency.

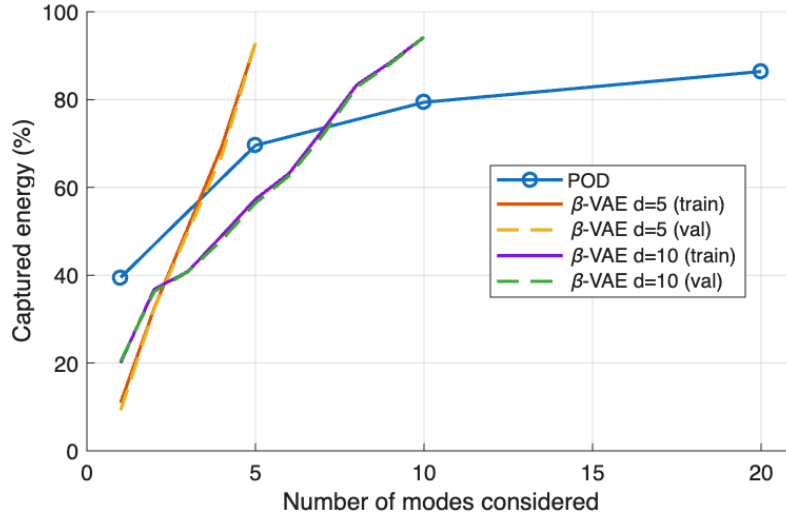


Figure 18: Captured energy (%) vs. number of modes for isotropic box turbulence.

4.3.4 Latent Space Disentanglement

Figure 19 presents the correlation matrices $|R_{ij}|$ of the latent space for the β -VAE ($\beta = 5 \times 10^{-4}$) and the POD temporal coefficients. The β -VAE correlation matrix exhibits a predominantly diagonal structure, with small off-diagonal values. Similarly as for the channel flow, but in this case for a latent dimension of 5, full orthogonality is not completely achieved, although the latent variables remain nearly uncorrelated.

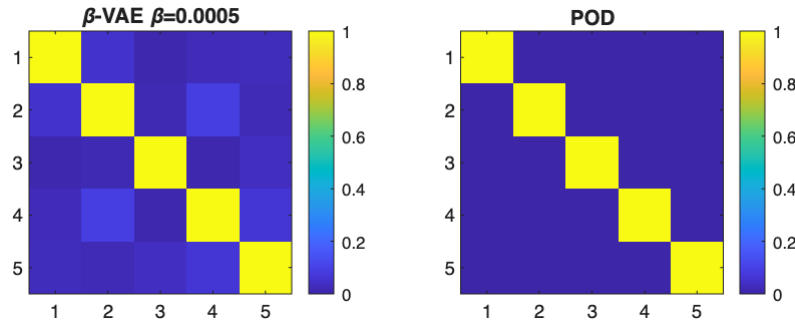


Figure 19: Absolute correlation matrices $|R_{ij}|$ of the latent representations for the homogeneous isotropic turbulence.

4.3.5 Turbulence Statistics

To assess whether the ROM reconstructions preserve turbulence physics beyond reconstruction accuracy, the energy spectrum and velocity fluctuation PDF are compared. Figure 20 shows the 1D streamwise energy spectrum $E(k_x)$ time-averaged over all snapshots. The raw DNS data follows the expected $k^{-5/3}$ scaling in the inertial range. The POD clearly deviated from the slope and the first wavenumber considered, but β -VAE reconstruct the spectrum well at low wavenumbers but underpredict energy at high wavenumbers. This behavior is expected from a ROM as it filters out small-scale information. Same behavior is observed in Halder et al. [6]. The β -VAE spectrum remains closer to the DNS reference across most of the inertial range, indicating better small-scale energy retention at the same number of modes.

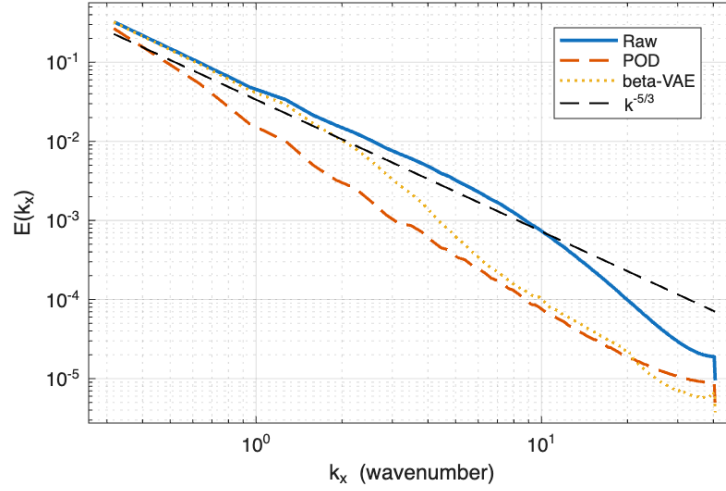


Figure 20: Time-averaged 1D energy spectrum $E(k_x)$ for box turbulence. The $k^{-5/3}$ Kolmogorov reference line is shown. Both ROMs underpredict the energy at high wavenumbers, with the β -VAE providing better fidelity than POD.

Figure 21 shows the probability density function (PDF) of the streamwise velocity fluctuations u' , normalized by the RMS and plotted on a semi-logarithmic scale to highlight the tails. The raw data exhibits approximately Gaussian tails; however, wider tails are expected due to the intermittency of turbulent flows. The β -VAE reconstruction captures the core, but is not able to capture the tails of the distribution. POD produces a narrower distribution due to the energy truncation and the absence of small-scale intermittent events.

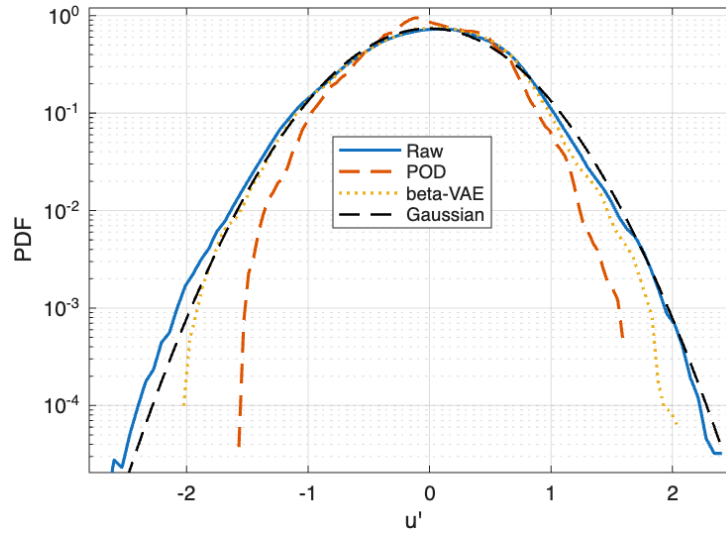


Figure 21: PDF of streamwise velocity fluctuations u' for box turbulence on a semi-logarithmic scale.

4.4 Transformer Temporal Prediction

The transformer is trained on the β -VAE latent time series of the cylinder flow ($Re = 100$, $d = 2$). The latent codes $\mathbf{z}(t) = [z_1(t), z_2(t)]$ trace a nearly circular orbit in 2D latent space, corresponding to the periodic vortex shedding cycle. The training set comprises the first 99% of the $N_t = 500$ snapshots (~ 495 steps) and the test set the last 1% (~ 5 steps).

Figure 22 shows the training and validation loss curves (MSE in normalized latent space) as a function of

epoch. The loss is clearly not decreasing and behaving in a noisy manner.

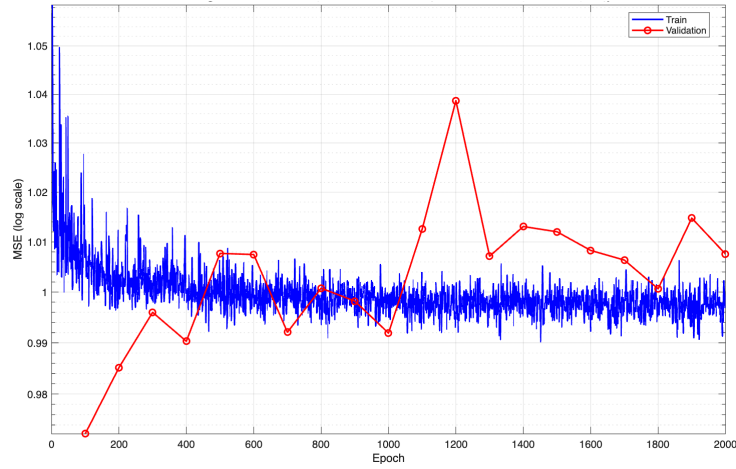


Figure 22: Training and validation MSE loss vs. epoch for the transformer trained on the cylinder β -VAE latent codes ($d = 2$).

Figure 23 shows the true and predicted latent trajectories $r_1(t)$ and $r_2(t)$ on the test set. It is clear by these figures that the transformer is not able to learn the dynamics of the flow in this case.

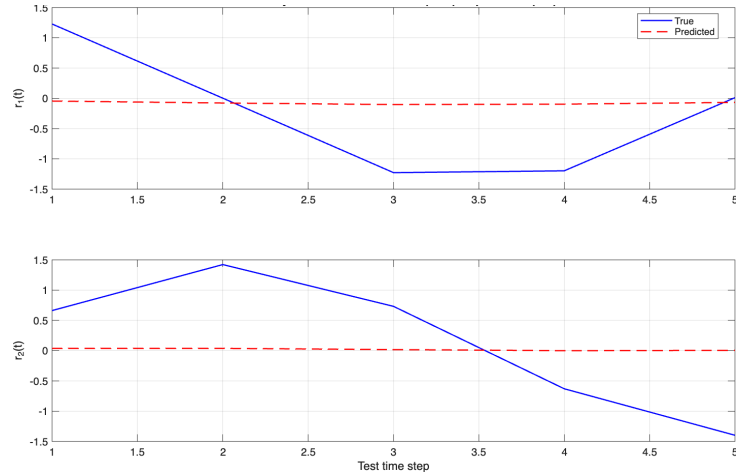


Figure 23: Transformer prediction on the cylinder test set. Left: true (blue) and predicted (red) latent trajectories $r_1(t)$, $r_2(t)$.

From this, it is inferred that the dataset used for the cylinder case, consisting of 500 snapshots, was not sufficient to achieve accurate temporal prediction. This conclusion is supported by the literature, where other studies report successful temporal prediction when using datasets containing more than 100,000 snapshots [14].



5 Discussion

5.1 Comparison of POD and β -VAE

The results consistently show that the β -VAE achieves higher energy compression than POD at equivalent or fewer modes across all three flow configurations. For the cylinder flow, both methods perform similarly because the dynamics are simple and are dominated by a single frequency and two spatially orthogonal modes that are as well represented by a linear basis as by a nonlinear one. The advantage of the β -VAE becomes key for the channel flow and box turbulence, where the nonlinear encoder can exploit correlations between flow structures at different scales that a linear basis cannot.

From the disentanglement perspective, the near diagonal correlation matrices confirm that the dominant modes are identified as nearly independent directions for the β -VAE. The β hyperparameter controls the trade-off between reconstruction fidelity and disentanglement: larger β produces a more diagonalized latent space but reduces the captured energy as proved in this work.

Figure 14 shows the inner-scaled mean velocity profile $U^+(y^+)$ together with the Reynolds stress distributions for the raw DNS data and the reconstructed fields. Both POD and β -VAE accurately reproduce the mean velocity profile, preserving the expected viscous sublayer and logarithmic region behavior. However, significant differences appear in the reconstruction of the Reynolds stresses as seen in Figure 15. While POD struggles to accurately capture the turbulent stress distributions, particularly in regions dominated by small-scale turbulent fluctuations, the β -VAE reconstruction remains much closer to the DNS reference. This indicates that the nonlinear latent representation learned by the β -VAE is more effective at preserving the turbulent dynamics and higher-order statistical information of the flow.

The inner-scaled channel flow results demonstrate that both POD and β -VAE are capable of accurately reconstructing the mean velocity profile, preserving the behavior of the turbulent boundary layer. However, important differences arise in the reconstruction of the Reynolds stresses. POD is unable to fully capture the turbulent stress distributions, whereas the β -VAE reconstruction remains very close to the DNS reference. These results suggest that, although both methods preserve the mean flow behavior, the nonlinear latent representation of the β -VAE is more effective at retaining the underlying turbulent dynamics of the channel flow.

Table 1: Comparison of Normalized Captured Energy (E) between POD and β -VAE across all flow configurations at matched latent dimensions.

Flow Configuration	Latent Dim. (r/z)	POD Energy	β -VAE Energy
Cylinder Wake ($Re = 100$)	3	98.5%	98.5%
Channel Flow ($Re_\tau = 1000$)	20	60.0%	96.0%
Isotropic Box ($Re_\lambda \approx 433$)	10	79.5%	94.0%

It is noteworthy that the performance discrepancy between POD and the β -VAE, while still significant, is less drastic in the isotropic turbulence case compared to the wall-bounded channel flow (Table 1). This behavior is fundamentally tied to the spatial statistics of the respective flow regimes. In the anisotropic channel flow, turbulent kinetic energy production and momentum transfer are driven by highly localized, non-linear bursting events (sweeps and ejections) near the wall. Because POD relies on global linear superposition, it structurally fails to resolve these sharp, localized gradients. Conversely, isotropic turbulence is statistically homogeneous in all directions, lacking boundary layers or mean shear. In a periodic homogeneous domain, POD modes mathematically degenerate toward global 3D Fourier modes [11]. Consequently, a global linear basis is marginally more effective at capturing the uniformly



distributed large-scale energy structures in isotropic turbulence, acting primarily as a standard low-pass filter. However, as evidenced by the premature truncation of the energy spectrum (Figure 20), POD still fundamentally fails to capture the non-linear transfer of energy to the dissipative micro-scales, an area where the β -VAE's non-linear manifold mapping remains strictly superior.

Regarding turbulence statistics, the energy spectrum comparison (Figure 20) reveals that both methods truncate the high-wavenumber energy, an expected consequence of dimensionality reduction, but the β -VAE retains more energy across the inertial range at the same number of modes. Similarly, the velocity fluctuation PDFs show that the β -VAE better captures the Gaussian of the turbulent fluctuations statistics that POD discards.

5.2 Spectral Analysis of Modal Temporal Coefficients

To evaluate the dynamical coherence of the extracted spatial structures, a frequency-domain analysis was performed on the temporal coefficients of the models—specifically, the right singular vectors $a_i(t)$ from POD and the latent variables $z_i(t)$ from the β -VAE. Following the analytical framework established in Spectral Proper Orthogonal Decomposition (SPOD) literature by Sieber et al. [13], taking the Fourier transform of the temporal coefficients allows us to explicitly link spatial modes to their driving physical timescales.

For the strictly periodic cylinder wake ($Re = 100$), the flow dynamics are dominated by a low-dimensional limit cycle. Consequently, standard spatial POD and the β -VAE both naturally extract modes that exhibit high temporal coherence. As seen in the frequency spectra alongside the spatial contours (Figure 7), the dominant modes for both methods exhibit a singular, sharp harmonic peak at a dimensionless frequency of $f \approx 0.17$. This strictly aligns with the fundamental shedding frequency (Strouhal number) of the von Kármán vortex street, demonstrating that both algorithms successfully disentangled the primary physical mechanism.

In contrast, as the flow transitions to wall-bounded and isotropic turbulence, the underlying physics become highly broadband. The frequency spectra for the channel (Figure 11) and box flow (Figure 17) modes exhibit spectral mixing, where energy is distributed across a wide range of frequencies rather than isolated harmonic peaks. This spectral analysis formally validates that while linear POD extracts modes purely based on variance, those high-energy spatial structures are driven by chaotic, multi-scale temporal dynamics characteristic of the turbulent energy cascade.

5.3 Limitations and Challenges

Several challenges were encountered in this work. First, training the β -VAE is significantly more complex than computing a POD: the neural network architecture requires careful hyperparameter selection (β , latent dimension d , learning rate, batch size), and convergence is not guaranteed. POD, by contrast, is obtained by a single SVD.

Second, the energy-vs-modes comparison in Section 4 measures actual reconstruction error with progressively more active latent dimensions, ordered by their temporal variance, rather than an analytical energy ranking. This makes the comparison meaningful but not strictly equivalent to the POD energy ordering.

Third, the transformer implementation faced several issues attributed to the quality of the datasets considered for this work.



6 Conclusions

This project rigorously compared POD and β -VAE as reduced-order modeling techniques across three distinct fluid dynamic regimes, evaluating both spatial compression efficiency and the preservation of turbulent statistics. The main conclusions are:

1. **β -VAE achieves superior compression efficiency for complex, non-linear flows.** For the highly turbulent cases, the β -VAE requires significantly fewer latent dimensions than POD to reach equivalent reconstruction accuracy. For instance, in the isotropic turbulence dataset at $z = 10$, the β -VAE captured 94.0% of the kinetic energy compared to POD's 79.5%. The non-linear activation functions within the encoder successfully map multi-scale correlations and curved turbulent manifolds that orthogonal linear hyperplanes cannot fully capture.
2. **POD remains computationally optimal for linear, periodic dynamics.** Both methods successfully identified the dominant physics of the periodic cylinder wake, converging to a 2D latent representation of the vortex shedding cycle. Both methods provided equivalent and highly accurate reconstructions (capturing 98.5% of the energy at rank 3). However, POD accomplishes this while entirely avoiding the extensive computational overhead required to train the neural network, proving that deep learning is unnecessary for simple laminar limit cycles.
3. **The β -VAE better preserves turbulence statistics.** The energy spectrum and velocity PDF comparisons for homogeneous isotropic turbulence show that the β -VAE reconstruction is closer to the DNS reference at high wavenumbers and better captures the Gaussian behavior of the fluctuation distribution. Similarly, for the turbulent channel flow, the β -VAE provides a more accurate reconstruction of the Reynolds stress distributions, demonstrating an improved capability to preserve turbulent statistics and small-scale turbulent dynamics.

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The reference implementation of the β -VAE and transformer framework follows the methodology of Solera-Rico et al. [14].



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7 Appendix A: Preliminary OpenFOAM Implementation and RANS Limitations

Prior to utilizing the high-fidelity Direct Numerical Simulation (DNS) datasets from the Johns Hopkins Turbulence Databases (JHTDB), preliminary numerical investigations were conducted by the authors using the open-source computational fluid dynamics (CFD) suite, OpenFOAM. The objective was to generate a custom dataset of a half-cylinder wake at a low Reynolds number to test the Reduced-Order Modeling (ROM) pipeline.

7.1 Simulation Setup

The simulation modeled the transient flow past a half-cylinder using a symmetry boundary condition along the centerline to reduce computational overhead. The incompressible, transient solver `pisoFoam` was utilized. To account for the coarse grid resolution, a Reynolds-Averaged Navier-Stokes (RANS) framework was applied, specifically utilizing the $k - \omega$ SST turbulence model to close the governing equations.

Table 2: OpenFOAM Simulation Specifications for Half-Cylinder Wake

Parameter	Specification
Software Suite	OpenFOAM v2206
Solver	<code>pisoFoam</code>
Turbulence Model	RANS ($k - \omega$ SST)
Reynolds Number (Re)	≈ 500
Time Step (Δt)	0.002 s (Courant Number ≤ 1.0)
Spatial Discretization	Second-order upwind
Temporal Discretization	Implicit Euler
Coarse Mesh Size	$\approx 3,200$ cells (Structured Hexahedral)
Fine Mesh Size	$\approx 60,000$ cells (256 x 256)
Boundary Conditions	Inlet (Velocity), Outlet (Pressure), Symmetry (Centerplane)

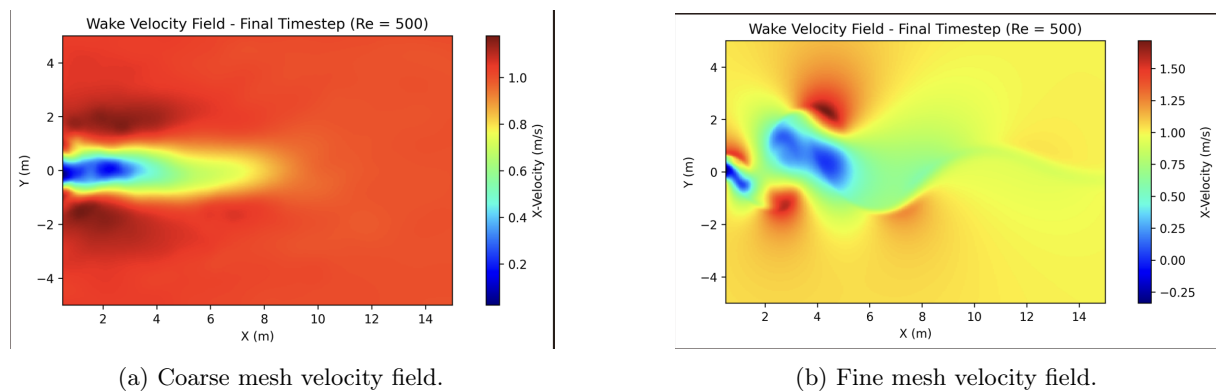


Figure 24: OpenFOAM $k - \omega$ SST RANS simulation results for the half-cylinder wake.

The domain was spatially discretized into a coarse and fine mesh to evaluate grid independence. Visual-



izations of the mesh resolution and the resulting velocity fields can be seen in Figure 24a (Coarse Data) and Figure 24b (Fine Data). The full simulation specifications are detailed in Table 2.

7.2 Fundamental Limitations of RANS for Data-Driven Models

When the OpenFOAM snapshot matrix was processed through the Proper Orthogonal Decomposition (POD) and β -Variational Autoencoder (β -VAE) pipelines, the resulting modal reconstructions were highly "blurry" and failed to capture distinct coherent structures. This failure is not a flaw in the deep learning architecture, but rather a fundamental mathematical incompatibility between RANS data and data-driven modal analysis [1].

Data-driven models require high spatial and temporal variance to map the underlying flow manifold. Using the Reynolds decomposition ($u = \bar{u} + u'$), algorithms like POD and VAEs rely on the instantaneous turbulent fluctuations (u') to identify orthogonal bases and latent features. However, RANS strictly solves for the time-averaged mean flow ($\bar{u}$). To close the equations without resolving the small scales, RANS utilizes the Boussinesq approximation, which models turbulent momentum transfer as an artificial increase in viscosity (eddy viscosity, ν_t).

This turbulent viscosity physically acts as a "blur filter," actively smearing sharp velocity gradients and artificially suppressing the dynamic variance of the von Kármán shedding [3]. Because the energy of the small scales is mathematically locked inside a scalar field (Turbulent Kinetic Energy, k) rather than being visible in the velocity kinematics, the ROM is effectively starved of the necessary dynamic range.

Other studies coupling OpenFOAM with DMD and POD have consistently demonstrated that capturing transient fluid kinematics requires scale-resolving simulations, such as Large Eddy Simulation (LES) or Direct Numerical Simulation (DNS) [9]. Consequently, to ensure the ROMs were evaluated on true physical manifolds containing highly non-linear momentum transfer, the project was successfully pivoted to utilize fully resolved DNS datasets.

8 Appendix B: Extraction of Temporal Dynamics via Dynamic Mode Decomposition (DMD)

While the primary objective of this report focuses on spatial compression methodologies (POD and β -VAE), a supplementary temporal analysis was performed on the laminar cylinder wake dataset ($Re = 100$) using Dynamic Mode Decomposition (DMD). This was conducted to validate that the raw snapshot matrix accurately captured the fundamental physical frequencies of the von Kármán vortex street prior to any reduced-order modeling.

8.1 Eigenvalue Spectrum and Stability

DMD is an equation-free, data-driven algorithm that extracts coherent structures coupled with single-frequency temporal growth/decay rates by approximating a linear operator $\mathbf{A}$ that steps the flow forward in time ($\mathbf{x}_{k+1} \approx \mathbf{A}\mathbf{x}_k$) [12].

The eigenvalues (λ) of this continuous-time operator dictate the stability of the extracted fluid structures. As shown in Figure 25, the eigenvalues of the cylinder wake are plotted on the complex plane relative to the unit circle.

- **On the Unit Circle ($|\lambda| = 1$):** The dominant modes sit exactly on the unit circle, indicating they

are neutrally stable (neither growing nor decaying). This mathematically confirms that the flow has reached a stable, periodic limit cycle.

- **Inside the Unit Circle ($|\lambda| < 1$):** Points lying inside the circle (such as the point near the negative real axis) represent highly damped transient modes. These are non-physical mathematical artifacts generated by the DMD algorithm to minimize the initial condition residuals [9], dissipating almost immediately.
- **Left Half-Plane:** Eigenvalues on the far left of the complex plane (near Real = -1) oscillate near the Nyquist frequency, flipping signs every timestep. These represent high-frequency numerical aliasing rather than macroscopic fluid structures.

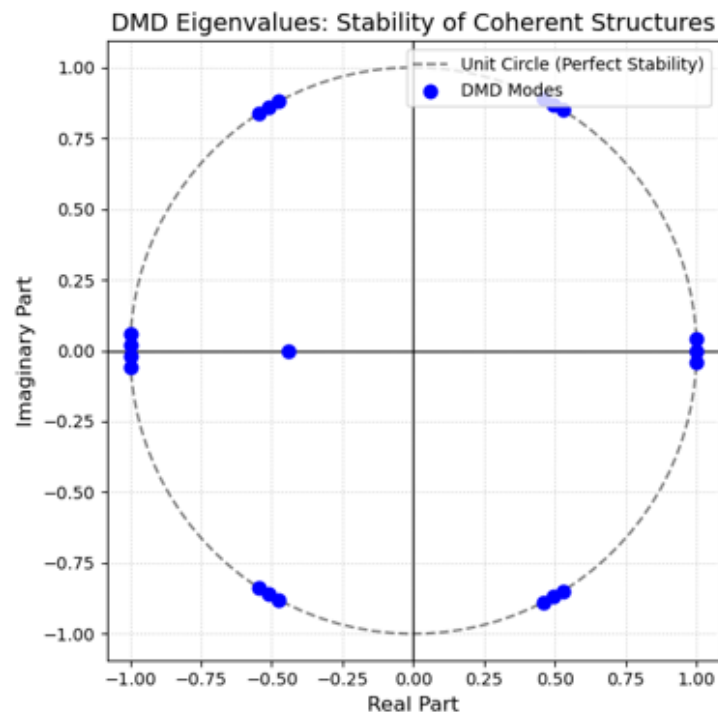


Figure 25: DMD eigenvalue spectrum. Dominant shedding modes are neutrally stable (residing on the unit circle).

8.2 Temporal Evolution and the Strouhal Number

The primary vortex shedding phenomenon is captured by a complex conjugate pair of eigenvalues located on the right half of the unit circle. The temporal evolution of these dominant modes (Modes 4 and 5) is plotted in Figure 26. The perfect sinusoidal overlap confirms a stable alternating shedding mechanism.

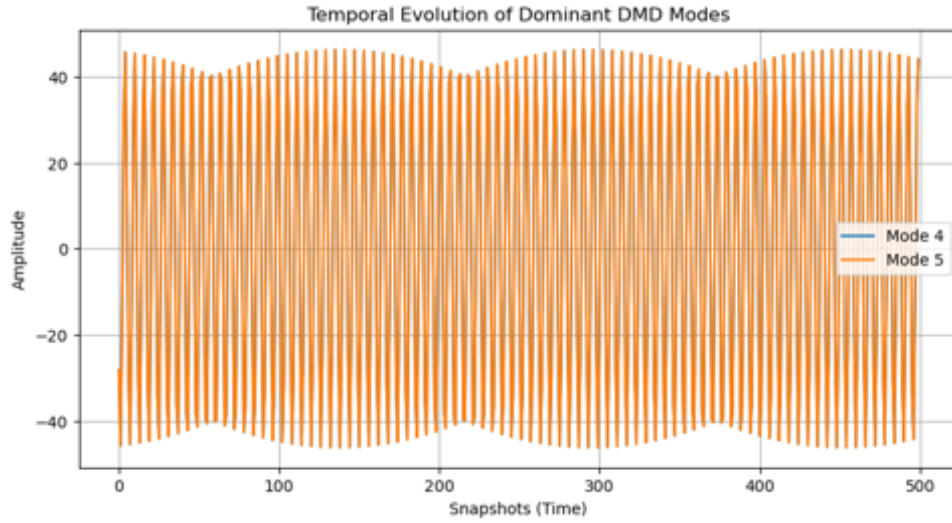


Figure 26: Temporal evolution of the dominant DMD conjugate pair (Modes 4 and 5), representing the primary vortex shedding cycle.

By extracting the angular frequency of this conjugate pair, we can calculate the dimensionless Strouhal number (St), which represents the ratio of unsteadiness to forward motion in the wake:

$$St = \frac{fL}{U} \quad (8)$$

where f is the shedding frequency, L is the characteristic length (cylinder diameter, D), and U is the free-stream velocity. Because the underlying DNS dataset is non-dimensionalized ($D = 1, U = 1$, with a normalized timestep spacing), the extracted dimensionless frequency directly yields the Strouhal number [18].

The DMD algorithm successfully isolated a dominant fundamental frequency corresponding to $St = 0.1675$. This aligns perfectly with the established theoretical and experimental Strouhal number for a circular cylinder at $Re = 100$ ($St \approx 0.164 - 0.168$), mathematically validating the temporal fidelity of the raw dataset [18].